

DSA0602 – DATA HANDLING AND VISUALIZATION

Slot C | CO1 – AT2

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Q1. Faceting / Small Multiples [R]

1. Write an R program using `facet_wrap()` to create a faceted plot showing the average-score trend for each Subject separately.

The following R program creates a separate panel for each subject and shows the average-score trend from Semester 1 to Semester 3.

```
library(ggplot2)

data <- data.frame(
  Subject = c("Maths","Maths","Maths",
             "Physics","Physics","Physics",
             "English","English","English",
             "CS","CS","CS"),
  Semester = c("Sem1","Sem2","Sem3",
              "Sem1","Sem2","Sem3",
              "Sem1","Sem2","Sem3",
              "Sem1","Sem2","Sem3"),
  AvgScore = c(72,78,81,65,70,75,80,82,85,74,83,90)
)

ggplot(data, aes(x = Semester, y = AvgScore, group = 1)) +
  geom_line() +
  geom_point(size = 3) +
  facet_wrap(~Subject) +
  labs(title = "Average Score Trend by Subject",
       x = "Semester", y = "Average Score") +
  theme_minimal()
```



2. Choose an appropriate variable to facet by and justify the choice.

Subject is the appropriate variable for faceting because it separates the data into four meaningful groups: Maths, Physics, English and CS. Each panel then shows the score trend for one subject independently.

3. Compare trends across the facets.

All subjects show an increasing trend from Semester 1 to Semester 3. Maths increases from 72 to 81, Physics from 65 to 75, English from 80 to 85, and CS from 74 to 90. CS shows the largest improvement, while English shows the smallest increase.

4. Explain the advantage of a single combined plot for this dataset.

A single combined plot places all subjects in one coordinate system, making direct comparison easier. For example, Semester 3 scores can be compared immediately: Maths 81, Physics 75, English 85 and CS 90.

5. Suggest a scenario where faceting would not be suitable.

Faceting is not suitable when there are too many categories because it produces many small panels and makes the visualization crowded. For example, creating one facet for 100 subjects or thousands of products would be difficult to read.

Q2. Highlighting with Color [R]

1. Write an R program to create a bar chart highlighting the expense category with the highest monthly spend.

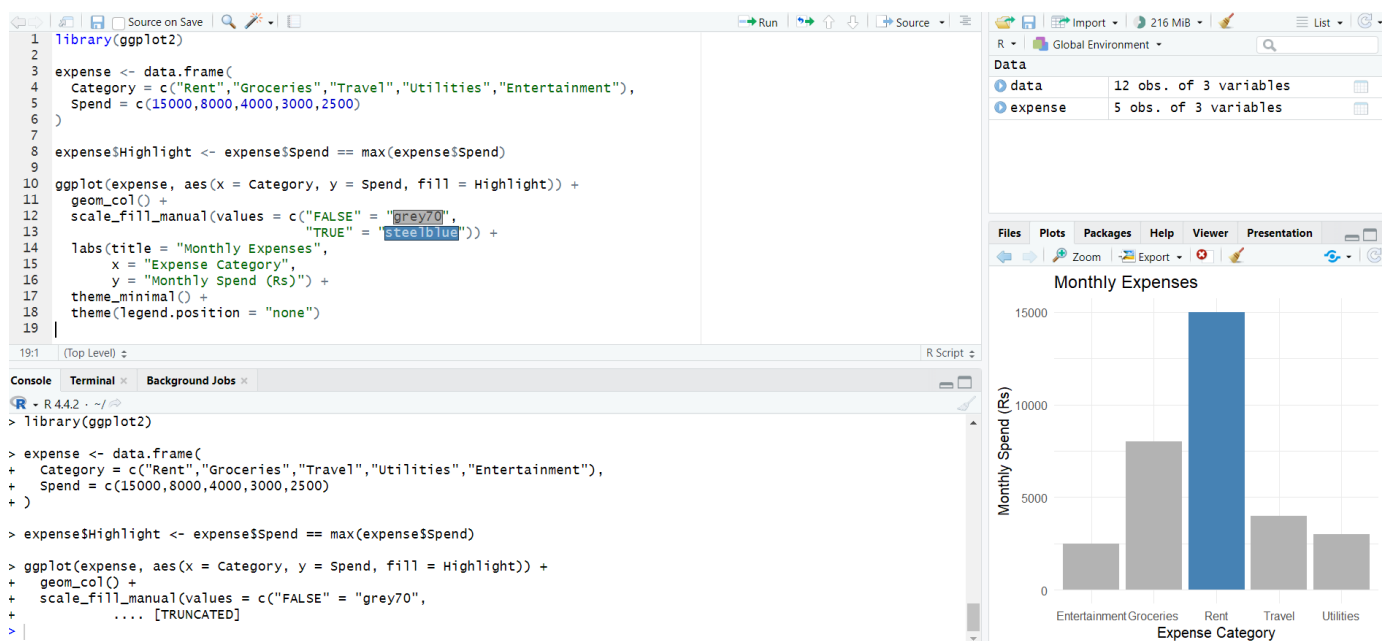
The program identifies the maximum spending category and highlights it with a different color while displaying the remaining categories in grey.

```
library(ggplot2)

expense <- data.frame(
  Category = c("Rent", "Groceries", "Travel", "Utilities", "Entertainment"),
  Spend = c(15000, 8000, 4000, 3000, 2500)
)

expense$Highlight <- expense$Spend == max(expense$Spend)

ggplot(expense, aes(x = Category, y = Spend, fill = Highlight)) +
  geom_col() +
  scale_fill_manual(values = c("FALSE" = "grey70",
                              "TRUE" = "steelblue")) +
  labs(title = "Monthly Expenses",
       x = "Expense Category",
       y = "Monthly Spend (Rs)") +
  theme_minimal() +
  theme(legend.position = "none")
```



2. Explain how highlighting improves data interpretation.

Highlighting directs the viewer's attention to the most important data point. Here, Rent is highlighted because it has the highest monthly spending. This allows the viewer to identify the key value quickly without examining every bar in detail.

3. Identify the highest-spending category.

Rent is the highest-spending category with a monthly expense of ₹15,000.

4. Discuss how excessive use of colors can affect interpretation.

Using too many bright colors can make a chart confusing and distracting. It can also make categories appear more important than they actually are. In addition, some color combinations are difficult for people with color-vision deficiencies. Color should therefore be used selectively.

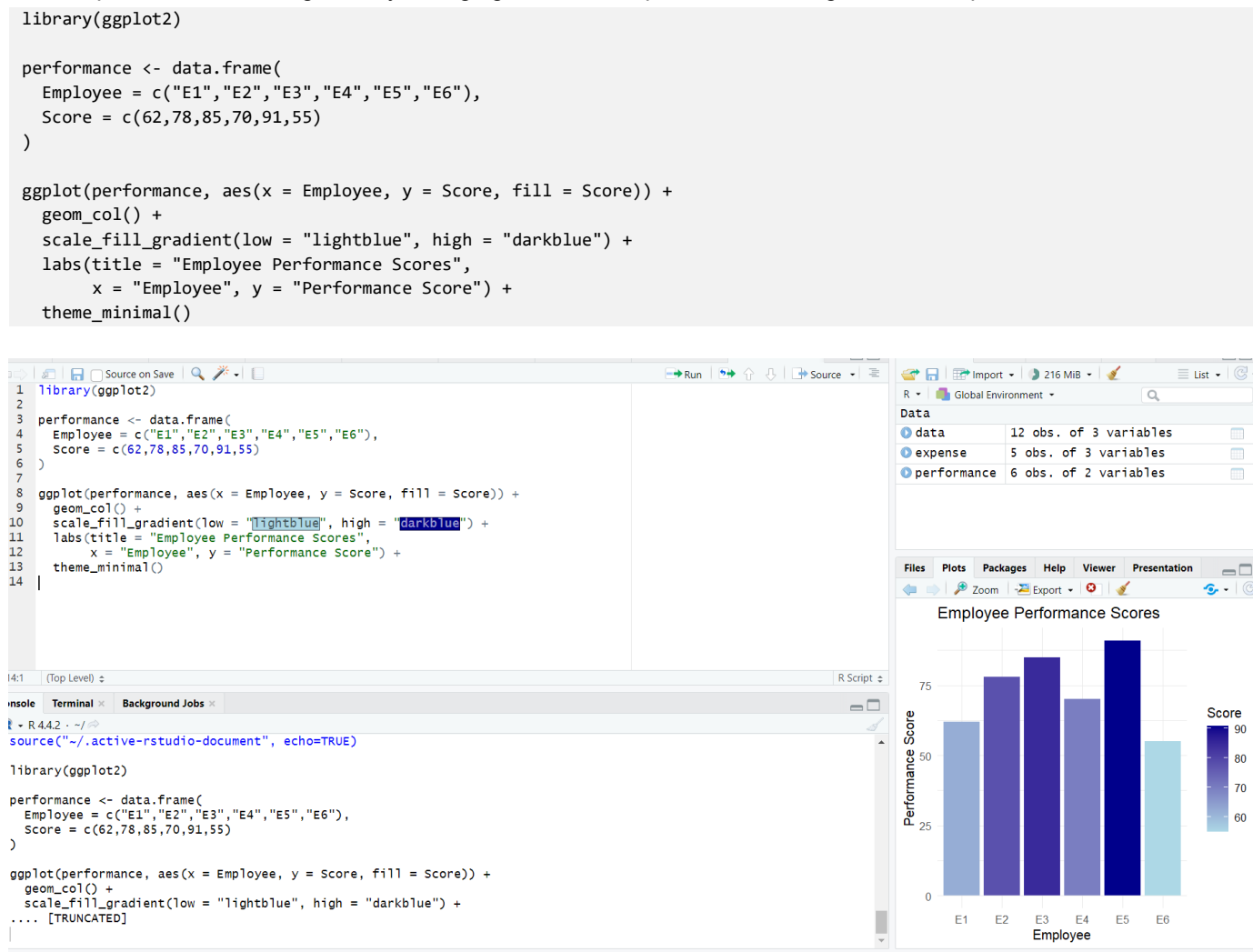
5. Suggest one alternative highlighting technique other than color.

Data labels or annotations can be used instead of color. For example, the highest value can be labelled as "Highest Expense – ₹15,000". Other alternatives include arrows, symbols, patterns or changing the bar size.

Q3. Color Scales for Data Representation [R]

1. Write an R program to create a bar chart using a sequential color scale based on Performance Score.

A sequential scale uses gradually changing shades to represent low-to-high numerical performance.



2. Identify the employees with the highest and lowest performance scores.

The highest performance is E5 with a score of 91. The lowest performance is E6 with a score of 55.

3. Explain why a sequential color scale is appropriate.

Performance Score is numerical and ordered from low to high. A sequential color scale represents this progression naturally: lighter shades can indicate lower scores and darker shades can indicate higher scores.

4. Suggest a color palette suitable for color-blind users.

A blue-based sequential palette is a suitable choice. The viridis palette is another strong option because it is designed for good perceptual separation and accessibility. Example: `scale_fill_viridis_c()`.

5. Explain the difference between sequential and categorical color scales.

Sequential color scales are used for ordered numerical data and progress gradually from one shade to another. Categorical scales use distinctly different colors to represent separate groups. For example, performance scores use a sequential scale, while departments such as HR, Finance and IT use a categorical scale.

Q4. Legends and Annotations [R]

1. Write an R program to plot the monthly stock-price trend for all three companies on a single chart with a clear legend.

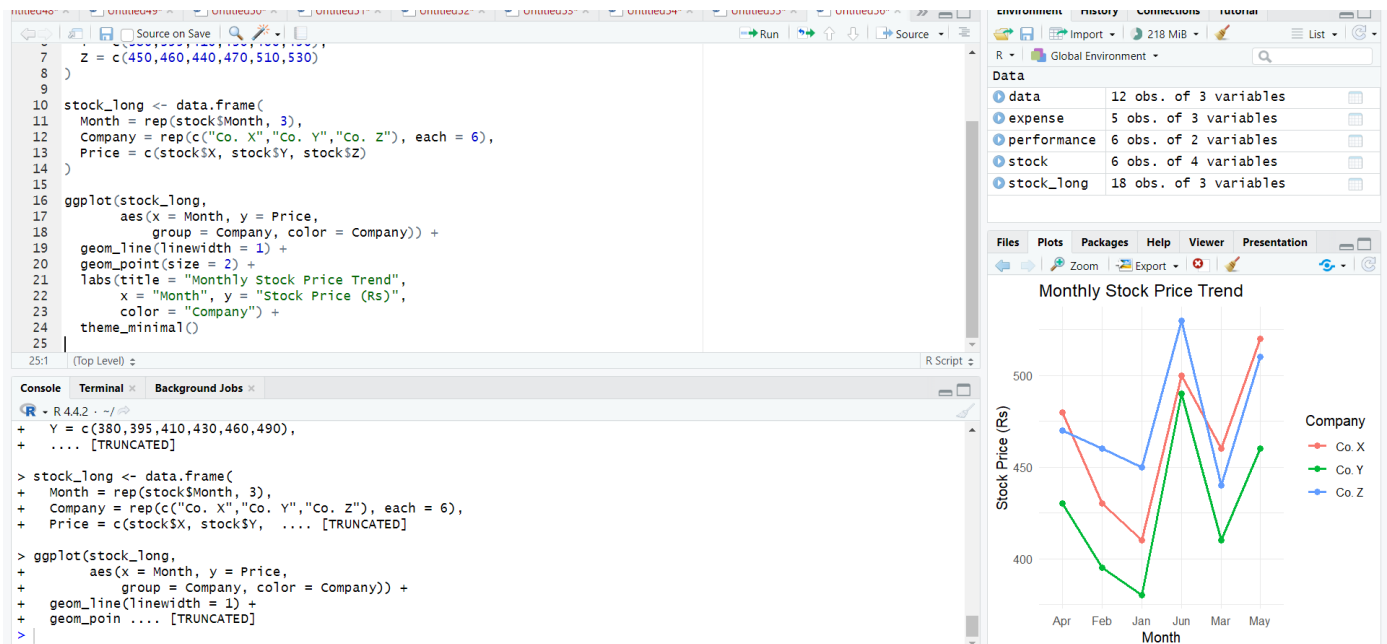
The data is reshaped into long format so that Company can be mapped to color and all three stock-price trends can be displayed on one chart.

```
library(ggplot2)

stock <- data.frame(
  Month = c("Jan", "Feb", "Mar", "Apr", "May", "Jun"),
  X = c(410, 430, 460, 480, 520, 500),
  Y = c(380, 395, 410, 430, 460, 490),
  Z = c(450, 460, 440, 470, 510, 530)
)

stock_long <- data.frame(
  Month = rep(stock$Month, 3),
  Company = rep(c("Co. X", "Co. Y", "Co. Z"), each = 6),
  Price = c(stock$X, stock$Y, stock$Z)
)

ggplot(stock_long,
  aes(x = Month, y = Price,
      group = Company, color = Company)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  labs(title = "Monthly Stock Price Trend",
       x = "Month", y = "Stock Price (Rs)",
       color = "Company") +
  theme_minimal()
```



2. Annotate (using annotate()/geom_text()) the point of highest price for each company.

The highest prices are Co. X = ₹520 in May, Co. Y = ₹490 in June, and Co. Z = ₹530 in June. These points can be labelled directly on the graph using annotate() or geom_text().

```
# Example annotation for Co. X
+ annotate("text", x = "May", y = 520,
  label = "Co. X Highest", vjust = -1)

# Similar annotations can be added for Co. Y and Co. Z.
```

3. Explain the role of legends in multi-series visualizations.

A legend explains which visual element corresponds to each data series. In this chart, it identifies the three lines as Co. X, Co. Y and Co. Z. Without a legend, it would be difficult to determine which line represents which company.

4. Discuss best practices for legends to avoid obscuring data.

Use clear labels, a meaningful legend title, distinct accessible colors, and a suitable position. The legend should not cover important data. It can be placed below the chart using `theme(legend.position = "bottom")`.

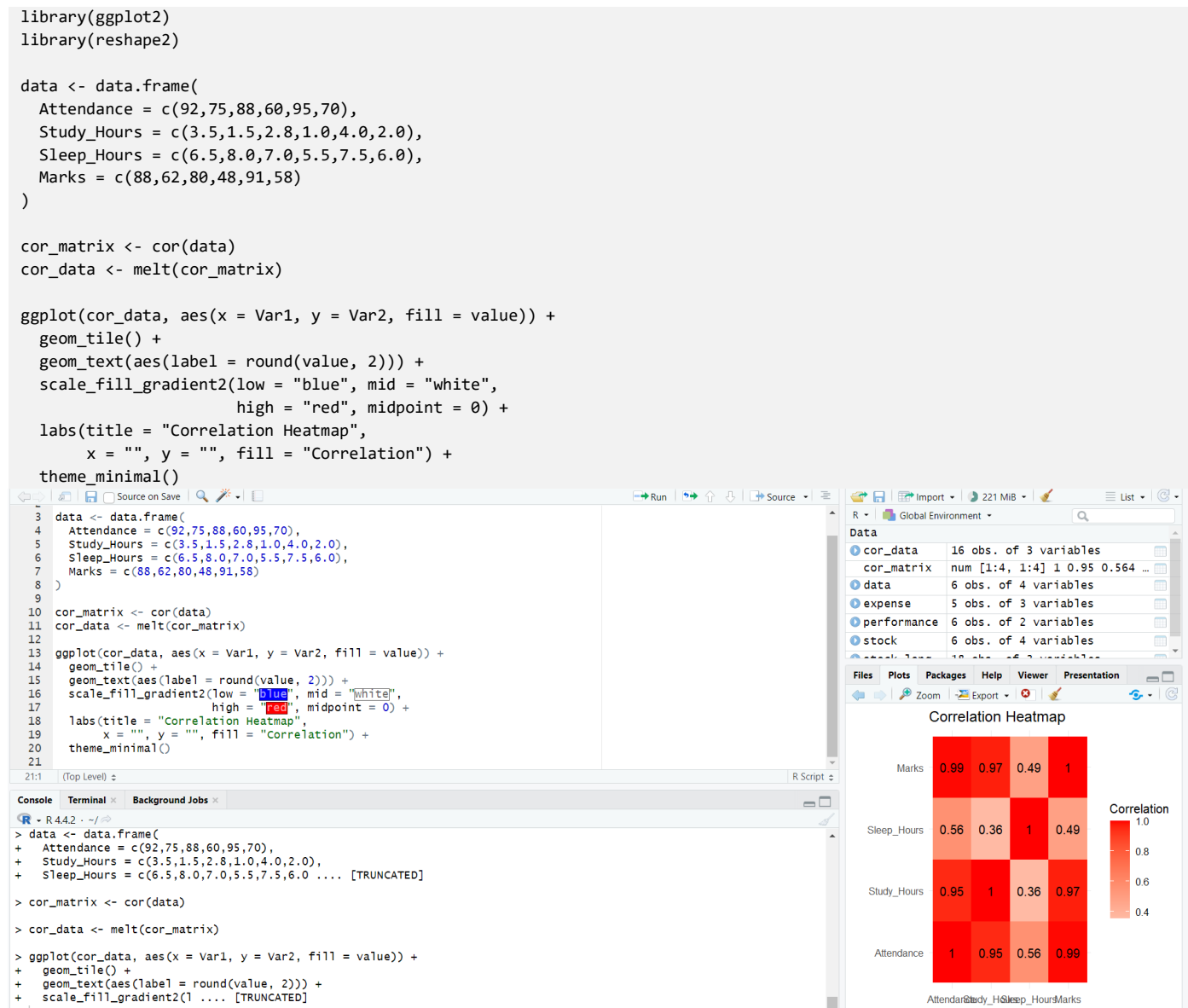
5. Suggest an alternative to a legend for distinguishing categories.

Direct labeling can replace a legend. The company name can be written next to its corresponding line. This reduces the need for the viewer to move their eyes between the chart and a separate legend.

Q5. Correlation Heatmap [R]

1. Write an R program to compute the correlation matrix of Attendance, Study Hours, Sleep Hours, and Marks, and display it as a heatmap.

The program calculates Pearson correlations using `cor()` and converts the matrix to long format for display using `geom_tile()`.



2. Identify the pair of variables with the strongest correlation.

The strongest correlation is between Attendance and Marks, with $r \approx 0.994$. This indicates a very strong positive linear relationship in this small dataset. Study Hours and Marks also show a strong positive relationship.

3. Explain what a diverging color scale communicates in a correlation heatmap.

A diverging color scale represents both positive and negative correlations. Values near +1 indicate a strong positive relationship, values near -1 indicate a strong negative relationship, and values near 0 indicate little or no linear relationship.

4. Discuss one limitation of using correlation to interpret this data.

Correlation does not prove causation. A strong correlation between attendance and marks does not by itself prove that increasing attendance causes higher marks. Also, the dataset contains only six students, so the result should not be generalized to a larger population without more data.

5. Suggest one additional variable that could improve this analysis.

Previous Academic Performance or previous-semester GPA could improve the analysis because it may explain variation in current marks alongside attendance, study time and sleep. Another useful variable could be study quality or study method.

Quick Exam Revision

Question	Main Concept	Key Point
Q1	Faceting	<code>facet_wrap(~Subject)</code>
Q2	Highlighting	Highlight Rent – ₹15,000
Q3	Sequential Colors	Highest E5 = 91; Lowest E6 = 55
Q4	Legends & Annotations	Highest: X = 520, Y = 490, Z = 530
Q5	Correlation Heatmap	Strongest: Attendance–Marks ≈ 0.994

Important R functions: `facet_wrap()`, `geom_col()`, `scale_fill_manual()`, `scale_fill_gradient()`, `scale_fill_gradient2()`, `geom_line()`, `geom_point()`, `labs()`, `annotate()`, `cor()`, `melt()`, `geom_tile()`.